DESIGN AND ANALYSIS OF ALGORITHMS · DAA

Master Theorem in Algorithm Analysis

PPT MADE USING GAMMA

PRESENTED BY

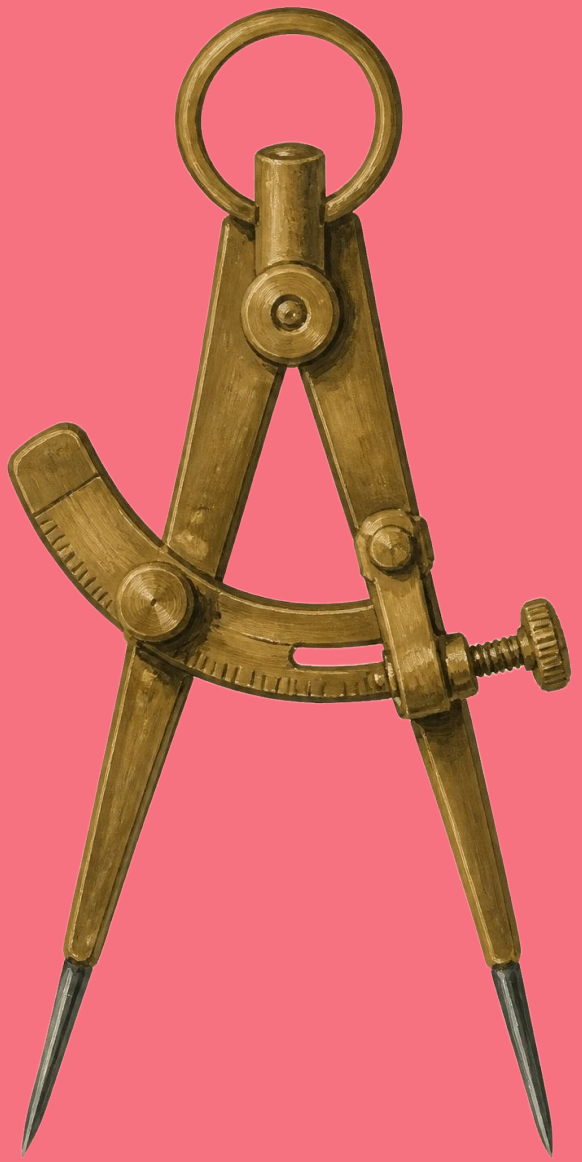
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ACADEMIC DETAILS

Sem-III (C) · CSE

SIT, Nagpur



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CORE PURPOSE

Solving divide-and-conquer recurrences

01 Solves recurrence relations directly

Avoids unrolling long algebraic expansions or drawing full recursion trees manually.

02 Determines asymptotic time complexity

Yields tight asymptotic bounds Θ for standard divide-and-conquer algorithms.

03 Simplifies recursive algorithm analysis

Provides an instant cookbook method for standard benchmarks like Merge Sort and Binary Search.



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STANDARD FORMULATION

General recurrence formula

RECURRENCE RELATION

$$T(n) = aT(n/b) + f(n)$$

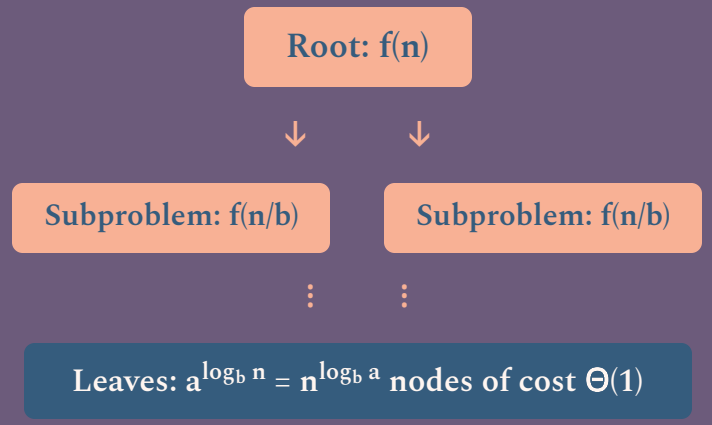
$a \geq 1$: number of recursive subproblems

n/b $(b > 1)$: size of each subproblem

$f(n)$: cost of dividing and combining solutions

$T(n)$: total time complexity on input size n

RECURSION TREE DECOMPOSITION



Total depth = $\log_b n$ levels

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CRITICAL COMPARISON

Three cases of the Master Theorem

Let p = log_b a. Here ε > 0 and k ≥ 0.

CASE 1

The Leaf Work Dominates

Condition: $f(n) = O(n^{\log_b a - \epsilon})$ for some constant ε > 0; non-recursive work f(n) is polynomially smaller than n^{log_b a.

Result: $T(n) = \Theta(n^{\log_b a})$

Most of the algorithm's time is spent at the bottom leaf level of the recursion tree. [1, 2, 3, 4]

CASE 2

The Work is Balanced Across Levels

Condition: $f(n) = \Theta(n^{\log_b a} \log^k n)$ for some constant k ≥ 0.

Result: $T(n) = \Theta(n^{\log_b a} \log^{k+1} n)$

Work is distributed evenly across recursion-tree levels; multiply the work at a single level by tree height (≈ log n). [1, 2, 3]

CASE 3

The Root / Non-Recursive Work Dominates

Condition: $f(n) = \Omega(n^{\log_b a + \epsilon})$ for some constant ε > 0, and the regularity condition: $af(n/b) \leq cf(n)$ for some constant c < 1, for all sufficiently large n. Master Theorem: Practice Problems and Solutions.

Result: $T(n) = \Theta(f(n))$

Most algorithm time is spent at the root level doing f(n) outside the recursive calls. [1, 2, 3]

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STEP-BY-STEP DERIVATION

Worked example: Merge Sort

$$T(n) = 2T(n/2) + n$$

1. Extract parameters: $a = 2,\ b = 2,\ f(n) = n$

2. Compute critical exponent: $\log_b a = \log_2 2 = 1$

3. Compare terms: $n^{\log_b a} = n^1 = n$

4. Match case: $f(n) = \Theta(n^1 \log^0 n)$ (Case 2 with $k = 0$)

Final Complexity: $T(n) = \Theta(n \log n)$

MERGE SORT RECURSION TREE

Level	Work
Level 0	Root size n; cost n
Level 1	2 subproblems of size $n/2$; $2 \times (n/2) = n$
Level 2	4 subproblems of size $n/4$; $4 \times (n/4) = n$
$\log_2 n$	n leaves of cost $\Theta(1)$; $n \times \Theta(1) = n$

Total Work = (Cost per Level: n) × (Tree Height: $\log_2 n$) = $\Theta(n \log n)$

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PRACTICAL SCOPE

Applications and limitations

PRIMARY APPLICATIONS

Merge Sort

$T(n) = 2T(n/2) + \Theta(n) \implies \Theta(n \log n)$

Binary Search

$T(n) = T(n/2) + \Theta(1) \implies \Theta(\log n)$

Strassen's Matrix Multiplication

$T(n) = 7T(n/2) + \Theta(n^2) \implies \Theta(n^{\log_2 7}) \approx \Theta(n^{2.81})$

KNOWN LIMITATIONS

Non-constant partitioning

Does not apply when subproblem sizes are unequal (e.g., $T(n) = T(n/3) + T(2n/3) + n$ in Quick Sort).

Polynomial gaps fail

Gaps must be strictly polynomial: cannot solve $T(n) = 2T(n/2) + n/\log n$.

Alternative techniques required

Must fall back to substitution methods, recursion trees, or the Akra-Bazzi method.

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TAKEAWAYS & WRAP-UP

Conclusion

Standardized recurrence analysis

Replaces tedious inductive algebraic work with a straightforward parameter comparison.

Categorized problem taxonomy

Classifies recurrence behavior into leaf-dominated, balanced, or root-dominated cases.

Essential engineering tool

Enables rapid estimation of asymptotic efficiency and architectural scalability.

Thank you!



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